Ankit Mathanker

Postdoctoral Scholar
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Molecular simulation and scientific machine learning for electrochemistry and catalysis.
Current focus: covalent organic framework cathodes for lithium-ion batteries.

Research Experience

Postdoctoral Scholar Sep 2026–Present
University of Chicago / Argonne National Laboratory, USA
Advisors: Prof. Laura Gagliardi (University of Chicago) and Dr. Rajeev Assary (Argonne National Laboratory)

- Developing covalent organic framework cathodes to improve the stability and capacity of lithium-ion batteries.

Graduate Research Assistant Aug 2021–Aug 2026
University of Michigan, Ann Arbor, USA
Advisor: Prof. Bryan R. Goldsmith

- Investigated how electrode potential, electrolyte ions, and co-reactants influence organic adsorption and electrocatalytic hydrogenation using density functional theory and molecular dynamics.
- Contributing to TSAgent, an agentic workflow for autonomous transition-state searches combining multi-agent planning, simulation automation, and density functional theory.

Graduate Intern Jun–Aug 2024
Lawrence Livermore National Laboratory, USA

- Estimated potential-dependent properties of metal–electrolyte interfaces using MACE machine learning interatomic potentials.

Graduate Research Assistant / Research Assistant Jan 2018–Jan 2021
University of Alberta, Edmonton, Canada

- Investigated hydrothermal liquefaction of agricultural and forest residues in supercritical water, including feedstock synergy and product characterization.

Undergraduate Dissertation Research May 2016–Apr 2017
IIT (ISM) Dhanbad, India

- Synthesized polyethyleneimine-impregnated adsorbents for CO_2 capture and modeled packed-bed adsorption in Aspen Adsorption.

Research Assistant, Summer Research Fellow May–Jul 2016
IIT Guwahati, India

- Synthesized low-cost tubular ceramic membranes by extrusion and characterized water flux and porosity.

Education

Ph.D., Chemical Engineering and Scientific Computing Aug 2021–Aug 2026
University of Michigan, Ann Arbor, USA

M.Sc., Chemical Engineering Jan 2018–Jan 2020
University of Alberta, Edmonton, Canada

B.Tech., Chemical Engineering Jul 2013–May 2017
IIT (ISM) Dhanbad, India

Publications

Peer-reviewed articles

1. **Mathanker, A.**; Guo, J.; Goldsmith, B. R.; Varley, J. B.; Govindarajan, N. Estimating potential-dependent physicochemical properties at metal–electrolyte interfaces using machine learning interatomic potentials. *ACS Electrochemistry* **2026**, 2(5), 1176–1189.
2. **Mathanker, A.**; Sharma, G.; Tran, B.; Singh, N.; Goldsmith, B. R. Effect of ions on the aqueous-phase adsorption of small aromatic organics on silver. *Journal of Physical Chemistry C* **2025**, 129(29), 13433–13444.
3. **Mathanker, A.**; Halarnkar, S.; Tran, B.; Singh, N.; Goldsmith, B. R. Synergistic effects in organic mixtures for enhanced catalytic hydrogenation and hydrodeoxygenation. *Chem Catalysis* **2024**, 4, 101135.
4. **Mathanker, A.**; Yu, W.; Singh, N.; Goldsmith, B. R. Effects of ions on electrocatalytic hydrogenation and oxidation of organics in aqueous phase. *Current Opinion in Electrochemistry* **2023**, 40, 101347.
5. Das, S.; **Mathanker, A.**; Pudasainee, D.; Khan, M.; Kumar, A.; Gupta, R. Synergistic effect of water and co-solvents on the hydrothermal liquefaction of agricultural biomass to produce heavy oil. *International Journal of Energy for a Clean Environment* **2022**, 23(4), 31–45.
6. **Mathanker, A.**; Das, S.; Pudasainee, D.; Khan, M.; Kumar, A.; Gupta, R. A review of hydrothermal liquefaction of biomass for biofuels production with a special focus on the effect of process parameters, co-solvents and extraction solvents. *Energies* **2021**, 14(16), 4916.
7. **Mathanker, A.**; Pudasainee, D.; Kumar, A.; Gupta, R. Hydrothermal liquefaction of lignocellulosic biomass feedstock to produce biofuels: Parameter study and products characterization. *Fuel* **2020**, 271, 117534.

Accepted / in press

8. de Kam, L. B. T.; Zhu, J.-X.; **Mathanker, A.**; Doblhoff-Dier, K.; Govindarajan, N. Benchmarking short-range machine learning potentials for atomistic simulations of metal/electrolyte interfaces. *Journal of Chemical Physics* **2026**, in press.

Preprints

9. Madhavan, V.; **Mathanker, A.**; Sweeney, D. M.; Ohiro, O. A.; Wang, Y.; Goldsmith, B. R. TSAgent: An Agentic Workflow for Autonomous Transition State Search. *arXiv:2605.14154* **2026**.

Theses

- **Ph.D. dissertation, University of Michigan, 2026.** Molecular-Level Insights into Organic Adsorption and Reactivity at Electrode-Electrolyte Interfaces.
- **M.Sc. thesis, University of Alberta, 2020.** Hydrothermal liquefaction of lignocellulosic biomass to produce biofuels.

Honors and Awards

- Visionary Award, RedoxFlow team, LLM Hackathon for Applications in Materials and Chemistry.
- Catalysis and Reaction Engineering Division Travel Award, AIChE Annual Meeting.
- Rackham Travel Award, University of Michigan, 2022, 2023, 2024.
- Captain Thomas Farrell Greenhalgh Memorial Graduate Scholarship, University of Alberta.
- Mary Louise Imrie Graduate Student Award, University of Alberta.
- Graduate Student Association Travel Award, University of Alberta.
- Summer Research Fellowship Programme Scholarship, Indian Academy of Sciences, Bengaluru.

Selected Presentations

- Estimating potential-dependent physicochemical properties at metal–electrolyte interface using machine learning interatomic potentials. *Gordon Research Conference: AI for Materials, Energy, and Chemical Science*, Texas, 2026.
- Estimating Water Coverage and Adsorption Isotherms at Electrolyte/Metal Interfaces Using Machine Learning Potentials. *AIChE Annual Meeting*, Boston, MA, 2025.
- Effect of ions on the aqueous-phase adsorption of small aromatic organics on silver. *AIChE Annual Meeting*, Boston, MA, 2025.
- Can aqueous ions modify the adsorption of organics on Ag? *Student and Postdoc Summer Seminar Series, CSiDIR*, University of Michigan, Ann Arbor, MI, 2025.
- Effect of ions on the aqueous-phase adsorption of small aromatic organics on Ag. *46th Michigan Catalysis Society Symposium*, Warren, MI, 2025.

Teaching and Academic Service

Teaching

- Graduate Teaching Assistant, CHE 538: Statistical Thermodynamics, University of Michigan, 2023.
- Graduate Teaching Assistant, CHE 316: Equilibrium Stage Processes, University of Alberta, Spring and Fall 2019.
- Graduate Teaching and Learning, Foundation and Practicum, University of Alberta, 2018–2019.

Professional and community service

- Co-chair, Gordon Research Seminar: AI for Materials, Energy and Chemical Sciences, Galveston, TX, 2028 (upcoming).
- Graduate Student Director, AIChE Catalysis and Reaction Engineering Division, 2025–2027.
- Graduate Student Representative, Michigan Chapter of the North American Catalysis Society, 2022–2024.
- Volunteer, UNITEA Community Social Work Team, University of Alberta, 2018–2019.
- Volunteer, Kartavya, IIT (ISM) Dhanbad, 2015–2017.

Technical Expertise

Atomistic modeling: Density functional theory; molecular dynamics; transition-state searches; microkinetic modeling; ASE, VASP, JDFTx, GROMACS, LAMMPS.

Scientific computing: Python, NumPy, SciPy, pandas, PyTorch; MACE interatomic potentials; active learning; agentic simulation workflows; Slurm and PBS.

Additional Experience

Management Trainee, Polyplex Corporation Ltd., India — Jul–Sep 2017